

Week2

```
CREATE DATABASE book_db;
```

```
USE book_db;
```

```
CREATE TABLE Books(
```

```
book_id INT PRIMARY KEY,
```

```
title VARCHAR(100) NOT NULL,
```

```
isbn VARCHAR(20) UNIQUE,
```

```
published_year INT CHECK(published_year<2027)
```

```
);
```

```
INSERT INTO Books(book_id, title, isbn, published_year) VALUES
```

```
(1, 'The Great Gatsby', '9780743273565', 1925),
```

```
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
```

```
(3, 'In The Cave', '9780451524935', 1949);
```

```
SELECT * FROM Books;
```

```
CREATE TABLE Members(
```

```
member_id INT PRIMARY KEY,
```

```
full_name VARCHAR(100),
```

```
email VARCHAR(100) UNIQUE
```

```
);
```

The screenshot shows a database management tool interface. The top section displays a list of SQL queries and their execution results. The bottom section shows a table of data.

Queries and Results:

#	Time	Action	Message	Duration
1	10:35:34	USE book_db	0 row(s) affected	0.000 sec
2	10:35:34	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn ...	Error Code: 1050. Table 'books' already exists	0.016 sec
3	10:35:44	INSERT INTO Books(book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '97...	Error Code: 1062. Duplicate entry '1' for key 'books.PRIMARY'	0.047 sec
4	10:35:49	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec
5	10:35:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email ...	Error Code: 1050. Table 'members' already exists	0.000 sec

Data Table:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945

```
INSERT INTO Members(member_id, full_name,email) VALUES
```

```
(101, 'Dikshita', 'dikshita@gmail.com'),
(102, 'Emma Wilson', 'emma.wilson@email.com'),
(103, 'Michael Brown', 'michael.brown@email.com');
```

```
SELECT * FROM Members;
```

The screenshot shows a database management tool interface. The top section displays SQL queries in a text editor. The bottom section shows the results of these queries in a table format. The table has columns for member_id, full_name, and email. The results show three members: John Smith, Emma Wilson, and Michael Brown. The interface also includes a sidebar with options like 'Result Grid', 'Form Editor', and 'Context Help'.

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

The bottom section shows the 'Output' pane with a list of actions and their messages. The actions include creating tables, inserting data, and selecting data. The messages indicate successful execution or errors.

#	Time	Action	Message
3	10:35:44	INSERT INTO Books(book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9...	Error Code: 1062. Duplicate entry '1' for key 'books.PRIMARY'
4	10:35:49	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned
5	10:35:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), ema...	Error Code: 1050. Table 'members' already exists
6	10:36:45	INSERT INTO Members(member_id, full_name, email) VALUES (101, 'Dikshita', 'dikshita@gm...	Error Code: 1062. Duplicate entry '101' for key 'members.PRIMARY'
7	10:36:45	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
8	10:36:45	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_da...	Error Code: 1050. Table 'loans' already exists

```
CREATE TABLE Loans(
loan_id INT PRIMARY KEY,
member_id INT,
book_id INT,
loan_date DATE,
FOREIGN KEY(member_id)REFERENCES Members(member_id),
FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

SELECT * FROM Loans;

The screenshot shows a database query editor with a SQL query and its result grid. The query is:

```
33 (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
34 (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
35 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
36 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
37 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
38 • SELECT * FROM Loans;
39 • SELECT m.full_name as Member_Name, b.title AS Book_Title
```

The result grid shows the following data:

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07

SELECT m.full_name as Member_Name, b.title AS Book_Title

FROM Loans l

INNER JOIN Members m ON l.member_id = m.member_id

INNER JOIN Books b ON l.book_id = b.book_id;

The screenshot shows a database query editor with a SQL query and its result grid. The query is:

```
37 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
38 • SELECT * FROM Loans;
39 • SELECT m.full_name as Member_Name, b.title AS Book_Title
40 FROM Loans l
41 INNER JOIN Members m ON l.member_id = m.member_id
42 INNER JOIN Books b ON l.book_id = b.book_id;
43 • SELECT published_year, COUNT(book_id) AS Total_Books
```

The result grid shows the following data:

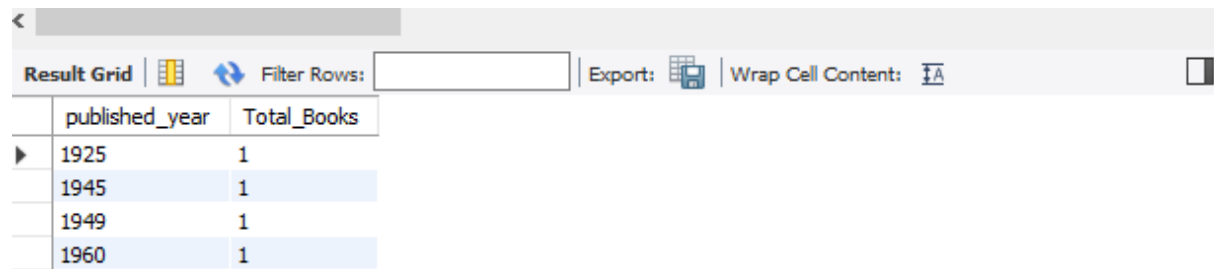
Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984

SELECT published_year, COUNT(book_id) AS Total_Books

FROM Books

GROUP BY published_year

ORDER BY published_year;



	published_year	Total_Books
▶	1925	1
	1945	1
	1949	1
	1960	1

```
CREATE TABLE Donation_History(  
    donation_id INT PRIMARY KEY,  
    book_id INT,  
    donor_name VARCHAR(100),  
    donation_date DATE,  
    FOREIGN KEY(book_id)REFERENCES Books(book_id)  
);  
  
START TRANSACTION;  
  
INSERT INTO Books (book_id, title, isbn, published_year)  
VALUES (4, 'Animal Farm', '9780451526342', 1945);  
  
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)  
VALUES (1, 4, 'Raj Kumar', CURDATE());  
  
COMMIT;  
  
CREATE INDEX idx_books_isbn ON Books(isbn);  
  
SELECT * FROM Books WHERE isbn = '9780451524935'
```

